General Properties of Many Particle Systems at Low Temperatures

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Contents

1. Second Quantisation

To note, second quantisation is used to track occupation number representation via creation/annihilation operators.

The occupation # tells you how many particles occupy each single particle energy level/state. We begin w/ the case of particles obeying Bose statistics. The wave function has the form:

$$\Phi_{N_1,N_2\ldots} = \underbrace{\left(\frac{N_1!N_2!\ldots}{N!}\right)^{1/2}}_{\text{Normalisation factor}} \underbrace{\sum_P}_{\substack{\text{permutation}\\ \text{sum}}} \underbrace{\varphi_{P_1}(\xi_1)\varphi_{P_2}(\xi_2)\ldots\varphi_{P_N}(\xi_N)}_{\text{individual particle states}} \tag{1}$$

Here the normalization factor satisfies:

$$\int |\Phi|^2 \prod_i d\xi_i = 1.$$

We let $F^{(1)}$ be an operator that is symmetric $\forall$ particles, $\therefore$ has the form

$$F^{(1)} = \sum_a f_a^{(1)} \tag{2}$$

we pick this b/c bosons (& fermions) are indistinguishable particles. f_a only acts on functions of ξ_a.

The corresponding diagonal matrix elements have the form:

$$\begin{aligned} \overline{F^{(1)}} &= \sum_i f_{ii}^{(1)} N_i \quad \text{, \& the off-diagonal elements} \\ \left(F^{(1)}\right)_{N_i,N_k}^{N_i-1,N_k+1} &= f_{ik}^{(1)}\sqrt{N_i N_k}, \end{aligned} \tag{3}$$

where

$$f_{ik}^{(1)} = \int \varphi_i^*(\xi) f^{(1)} \varphi_k(\xi) d\xi$$

From $F^{(1)}$, we can introduce creation & annihilation operators.

1.1. A Brief re–visit to Creation and Annihilation Operators

Starting w/ the Hamiltonian of the quantum harmonic oscillator:

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2$$

Factoring out a term of $\hbar\omega$, the Hamiltonian becomes:

$$\hat{H} = \hbar\omega \left[\frac{\hat{p}^2}{2m\hbar\omega} + \frac{1}{2} \frac{m\omega}{\hbar} \hat{x}^2 \right]$$

By re-expression, we have :

$$\hat{H} = \hbar\omega \left[\left(\frac{1}{\sqrt{2m\hbar\omega}} \hat{p} \right)^2 + \left(\sqrt{\frac{m\omega}{2\hbar}} \hat{x} \right)^2 \right]$$

Since this is representation of a sum of squares, we can re-express again as :

$$\begin{aligned} a &= \sqrt{\frac{m\omega}{2\hbar}} \hat{x} + i \frac{1}{\sqrt{2m\hbar\omega}} \hat{p} \\ &= \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) \end{aligned}$$

The Hermitian conjugate is this :

$$a^\dagger = \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right)$$

When we compute $a^\dagger a$, we get the number operator :

$$\begin{aligned} a^\dagger a &= \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \sqrt{\frac{m\omega}{2\hbar}} \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) \\ &= \frac{m\omega}{2\hbar} \left[\left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) \right] \end{aligned}$$

Computing the bracketed term,

$$\begin{aligned} &\left(\hat{x} - \frac{i}{m\omega} \hat{p} \right) \left(\hat{x} + \frac{i}{m\omega} \hat{p} \right) \\ &= \hat{x}^2 + \frac{i}{m\omega} \hat{x}\hat{p} - \frac{i}{m\omega} \hat{p}\hat{x} + \hat{p}^2 \\ &= \frac{m\omega}{2\hbar} \hat{x}^2 + \frac{\hat{p}^2}{2m\hbar} - \frac{1}{2} \\ &= \frac{1}{\hbar\omega} \left(\frac{\hat{p}^2}{2m} + \frac{1}{2} m\omega^2 \hat{x}^2 \right) - \frac{1}{2} \end{aligned}$$

$$a^\dagger a = \frac{\hat{H}}{\hbar\omega} - \frac{1}{2}$$

By rearrangement, we can see that :

$$\hat{H} = \hbar\omega\left(a^\dagger a + \frac{1}{2}\right)$$

From this expression, we define $a^\dagger a$ as the number operator:

$$\hat{N} = a^\dagger a$$

If we take the commutator of a & $a^\dagger$,

$$\begin{aligned}
[a, a^\dagger] &= aa^\dagger - a^\dagger a \\
&= \frac{m\omega}{2\hbar}\left[\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)\left(\hat{x} - \frac{i}{m\omega}\hat{p}\right) - \left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)\right] \\
&= \frac{m\omega}{2\hbar}\left[\left(\hat{x}^2 - \frac{i}{m\omega}\hat{x}\hat{p} + \frac{i}{m\omega}\hat{p}\hat{x} - \frac{i^2}{m^2\omega^2}\hat{p}^2\right) - \left(\hat{x} - \frac{i}{m\omega}\hat{p}\right)\left(\hat{x} + \frac{i}{m\omega}\hat{p}\right)\right] \\
&= \frac{m\omega}{2\hbar}\left[\left(\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} - \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x})\right) - \left(\hat{x}^2 + \frac{i}{m\omega}\hat{x}\hat{p} - \frac{i}{m\omega}\hat{p}\hat{x} - \frac{i^2}{m^2\omega^2}\hat{p}^2\right)\right] \\
&= \frac{m\omega}{2\hbar}\left[\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} - \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) - \left(\hat{x}^2 + \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) + \frac{\hat{p}^2}{m^2\omega^2}\right)\right] \\
&= \frac{m\omega}{2\hbar}\left[\hat{x}^2 + \frac{\hat{p}^2}{m^2\omega^2} - \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) - \hat{x}^2 - \frac{i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x}) - \frac{\hat{p}^2}{m^2\omega^2}\right] \\
&= \frac{m\omega}{2\hbar}\left[\frac{-2i}{m\omega}(\hat{x}\hat{p} - \hat{p}\hat{x})\right] \\
&= \frac{m\omega}{2\hbar}\cdot\left(\frac{-2i}{m\omega}\right)(\hat{x}\hat{p} - \hat{p}\hat{x}) \\
&= -\frac{i}{\hbar}(\hat{x}\hat{p} - \hat{p}\hat{x}) = -\frac{i}{\hbar}[\hat{x}, \hat{p}] = -\frac{i}{\hbar}(i\hbar) = -i^2 = -(-1) = 1.
\end{aligned}$$

When acting on an eigenstate $|n\rangle$, they yield :

$$\begin{cases} a|n\rangle = \sqrt{n}|n-1\rangle \\ a^\dagger|n\rangle = \sqrt{n+1}|n+1\rangle \end{cases}$$

2. Picking up after the "Re–visit."

Similarly, we introduce the annihilation operator, which decreases by one the number of particles in the ith state & possess matrix elements :

$$(a_i)_{N_i}^{N_i - 1} = \sqrt{N_i} \tag{4}$$

To comment on the notation, $N_i - 1$ represents the final number of particles in state i, & N_i represents the initial number of particles in state i. This can further be re-expressed in Dirac notation as:

$$\langle N_i - 1|a_i|N_i\rangle = \sqrt{N_i}$$

The Hermitian conjugate, or creation operator increases the number of particles by one.

$$(a_i^\dagger)_{N_i-1}^{N_i} = \left((a_i)_{N_i}^{N_i-1}\right)^* = \sqrt{N_i} \tag{5}$$

By (3), the operator $F^{(1)}$ can be written as:

$$F^{(1)} = f_{ik}^{(1)} a_i^\dagger a_k \tag{6}$$

Eq. (6) is the expression for $F^{(1)}$ in the second quantisation form. By (4) & (5), we can compute the products of both the creation & annihilation operators.

$$a_i^\dagger a_i = \sqrt{N_i}\sqrt{N_i} = N_i$$

Although that may have seemed a bit too obvious, the product of $a_i a_i^\dagger$ is not the same.

$$a_i a_i^\dagger = (a_i)_{N_i}^{N_i-1}(a_i^\dagger)_{N_i-1}^{N_i}$$

From a_i, let the final states be $M = N_i - 1$. By rearrangement, $N_i = M + 1$. Note that M is a dummy variable. By this, the expression becomes:

$$a_i a_i^\dagger = (a_i^\dagger)_M^{M+1}(a_i)_{M+1}^M = \sqrt{M+1}\sqrt{M+1}$$

Replacing our dummy variable, we obtain:

$$(a_i)_{N_i}^{N_i-1}(a_i^\dagger)_{N_i-1}^{N_i} = \sqrt{N_i+1}\sqrt{N_i+1} = N_i + 1$$

In summary,

$$\begin{cases} a_i^\dagger a_i = N_i \\ a_i a_i^\dagger = N_i + 1 \end{cases} \tag{7}$$

By (4), (5), & (7), we can establish the following commutation relations.

$$[a_i, a_k^\dagger] = a_i a_k^\dagger - a_k^\dagger a_i$$

recall: $a_i = (a_i)_{N_i}^{N_i-1} = \sqrt{N_i}$. Let $a_k^\dagger$ take the form of (5), w/ an index of k:

$$a_k^\dagger = (a_k^\dagger)_{N_k-1}^{N_k} = \sqrt{N_k}$$

applying these creation & annihilation operators to the commutation relation, we obtain:

$$\begin{aligned} [a_i, a_k^\dagger] &= a_i a_k^\dagger - a_k^\dagger a_i \\ &= (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_{N_k-1}^{N_k} - (a_k^\dagger)_{N_k-1}^{N_k}(a_i)_{N_i}^{N_i-1} \end{aligned}$$

A helpful reminder before proceeding: operators act right to left. Thus, $(a_k^\dagger)$ only alters mode k. (a_i) only alters mode i. The nice thing about (a_i) is that we had originally defined it by (4).

Computing the first term,

$$a_i a_k^\dagger = (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_{N_k-1}^{N_k} = \sqrt{N_i}(a_k^\dagger)_{N_k-1}^{N_k}$$

For $(a_k^\dagger)_{N_k-1}^{N_k}$, we employ the use of dummy variable M. So again, we let $M = N_k - 1$. Thus, $N_k = M + 1$. Applying the dummy variables, we obtain (for the first term):

$$a_i a_k^\dagger = (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_M^{M+1} = \sqrt{N_i}\sqrt{M+1}$$

Replacing the dummy variable reveals the first term as:

$$a_i a_k^\dagger = (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_{N_k-1}^{N_k} = \sqrt{N_i}\sqrt{N_k+1}$$

Repeating the process for the second term,

$$a_k^\dagger a_i = (a_k^\dagger)_{N_k-1}^{N_k}(a_i)_{N_i}^{N_i-1}$$

Notably, $(a_k^\dagger)_{N_k-1}^{N_k}$ was previously defined in (5). Using the dummy variable substitution,

$$M = N_i - 1 \quad ; \quad N_i = M + 1$$

Thus,

$$a_k^\dagger a_i = (a_k^\dagger)_{N_k-1}^{N_k}(a_i)_{M+1}^{M} = \sqrt{N_k}\sqrt{M+1} = \sqrt{N_k}\sqrt{N_i+1}$$

Combining the terms of the commutation relation, we obtain:

$$\begin{aligned}
[a_i, a_k^\dagger] &= a_i a_k^\dagger - a_k^\dagger a_i \\
&= (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_{N_k-1}^{N_k} - (a_k^\dagger)_{N_k-1}^{N_k}(a_i)_{N_i}^{N_i-1} \\
&= \sqrt{N_i}\sqrt{N_k+1} - \sqrt{N_k}\sqrt{N_i+1}
\end{aligned}$$

Evidently, we can see that when $i \neq k$, $[a_i, a_k^\dagger] = 0$. Recomputing this commutation relation when $i = k$ reveals:

$$\begin{aligned}
[a_i, a_i^\dagger] &= a_i a_i^\dagger - a_i^\dagger a_i \\
&= (a_i)_{N_i}^{N_i-1}(a_k^\dagger)_{N_k-1}^{N_k} - (a_k^\dagger)_{N_k-1}^{N_k}(a_i)_{N_i}^{N_i-1} \\
&= (a_i)_{N_i}^{N_i-1}(a_i^\dagger)_{N_i-1}^{N_i} - (a_i^\dagger)_{N_i-1}^{N_i}(a_i)_{N_i}^{N_i-1}
\end{aligned}$$

By (7),

$$= (N_i + 1) - (N_i) = N_i + 1 - N_i = N_i - N_i + 1 = 1.$$

At this point, it becomes quite evident that:

$$[a_i, a_k^\dagger] = \delta_{ik}$$

Similarly, we can show that $[a_i, a_k] = [a_i^\dagger, a_k^\dagger] = 0$.

$$\begin{aligned}
[a_i, a_k] &= a_i a_k - a_k a_i \\
&= (a_i)_{N_i}^{N_i-1}(a_k)_{N_k}^{N_k-1} - (a_k)_{N_k}^{N_k-1}(a_i)_{N_i}^{N_i-1} \\
&= \sqrt{N_i}\sqrt{N_k} - \sqrt{N_k}\sqrt{N_i} = 0
\end{aligned}$$

As a brief sanity check:

$$\begin{aligned}
[a_i^\dagger, a_k^\dagger] &= a_i^\dagger a_k^\dagger - a_k^\dagger a_i^\dagger \\
&= (a_i^\dagger)_{N_i-1}^{N_i}(a_k^\dagger)_{N_k-1}^{N_k} - (a_k^\dagger)_{N_k-1}^{N_k}(a_i^\dagger)_{N_i-1}^{N_i} \\
&= \sqrt{N_i+1}\sqrt{N_k+1} - \sqrt{N_k+1}\sqrt{N_i+1} = 0
\end{aligned}$$

To summarize commutation relation,

$$[a_i, a_j^\dagger] = a_i a_j^\dagger - a_j^\dagger a_i = \delta_{ij} \tag{8}$$

$$[a_i, a_k] = [a_i^\dagger, a_k^\dagger] = 0$$

Similar representations are possible for symmetric operators:

$$F^{(2)} = \sum_{a,b} f_{ab}^{(2)} \tag{9}$$

$f_{ab}^{(2)}$ acts on ξ_a & ξ_b. In the second quantization, (9) becomes:

$$F^{(2)} = \sum_{iklm} f_{lm}^{(2)ik} (a_i)^\dagger (a_k)^\dagger a_l a_m \tag{10}$$

where

$$f_{lm}^{(2)ik} = \int \varphi_i^*(\xi_1)\varphi_k^*(\xi_2) f^{(2)} \varphi_l(\xi_1)\varphi_m(\xi_2) d\xi_1 d\xi_2$$

In the second quantisation, particle symmetry is automatically absorbed & handled by creation/annihilation operators. Rather than looking at specific particles a & b, the perspective assumes that of the quantum states (i, k, l, m).

The Hamiltonian of a system of interacting particles situated in an external field:

$$H = \sum_a H_a^{(1)} + \sum_{a,b} U^{(2)}(\mathbf{r}_a, \mathbf{r}_b) + \sum_{a,b,c} U^{(3)}(\mathbf{r}_a, \mathbf{r}_b, \mathbf{r}_c) + \dots \tag{11}$$

Note $H_a^{(1)} = \left(-\frac{\hbar^2 \nabla_a^2}{2m}\right) + U(\mathbf{r}_a)$. Each term of (11) represents an N-body system where $N \in \mathbb{N}$ (excluding 0).

In the second quantisation,

$$H = \sum_{ik} H_{ik} a_i^\dagger a_k + \sum_{iklm} U_{lm}^{(2)ik} a_i^\dagger a_k^\dagger a_l a_m \dots \tag{12}$$

For single states φ_i, if we choose the eigenfunctions of the Hamiltonian $H_a^{(1)}$, the first term in (12) becomes:

$$H^{(1)} = \sum_i \mathcal{E}_i a_i^\dagger a_i = \sum_i \mathcal{E}_i N_i \tag{13}$$

Recall that is Fermi statistics, the complete wave function of a system is antisymmetric w/ respect to all the variables. This implies that occupation numbers must be only 0 or 1 (for non-interacting particles). Wavefunctions have the form:

$$\Phi_{N_1, N_2, \dots} = \frac{1}{\sqrt{N!}} \sum_P (-1)^P \varphi_{P_1}(\xi_1)\varphi_{P_2}(\xi_2) \dots \varphi_{P_N}(\xi_N) \tag{14}$$

where the numbers $P_1, P_2, \dots, P_N$ are different. The term $(-1)^P$ indicates that odd permutation (swapping particle positions) appear w/ a minus sign. For definiteness, we pick an ordering presentation such that:

$$P_1 < P_2 < \dots < P_N \tag{15}$$

For an operator $F^{(1)}$ of type (2), the matrix elements are:

$$\begin{aligned} &\text{i) Diagonal}: \ \overline{F^{(1)}} = \sum_i f_{ii} N_i \\ &\text{ii) Off-Diagonal}: \ (F^{(1)})_{0_i, 1_k}^{1_i, 0_k} = \pm f_{ik}^{(1)} \end{aligned} \tag{16}$$

Now, we introduce new annihilation/creation operators such that:

$$(a_i)_1^0 = (a_i^\dagger)_0^1 = (-1)^{\sum_{l=1}^{i-1} N_l} \tag{17}$$

The products of these new operators are found by letting $P = \sum_{l=1}^{i-1} N_l$. Let:

$$(a_i)_1^0 = (-1)^P$$
$$(a_i^\dagger)_0^1 = (-1)^P$$

$$a_i^\dagger a_i = (a_i^\dagger)_0^1 (a_i)_1^0 = (-1)^P(-1)^P = (-1)^{2P} = 1 = N_i \quad (\equiv 0 \text{ if starting w/ 0 particles})$$
$$a_i a_i^\dagger = (a_i)_1^0 (a_i^\dagger)_0^1 = (-1)^P(-1)^P = (-1)^{2P} = 1 = 1 - N_i \quad (\equiv 0 \text{ if starting w/ 1 particle})$$

$\therefore$ in sum:

$$\begin{cases} a_i^\dagger a_i = N_i \\ a_i a_i^\dagger = 1 - N_i \end{cases} \tag{18}$$

Computing the anticommutators $\{A, B\} \equiv AB + BA$:

$$\begin{aligned} \{a_i, a_i^\dagger\} &\equiv a_i a_i^\dagger + a_i^\dagger a_i \\ &= (1 - N_i) + N_i \\ &= 1 \end{aligned}$$

For distinct states $i \neq k$, without loss of generality assuming $i < k$, the phase string context alters by an odd permutation factor during cross-over actions, yielding:

$$\begin{aligned} a_i a_k^\dagger &= -a_k^\dagger a_i \implies \{a_i, a_k^\dagger\} = 0 \\ a_i a_k &= -a_k a_i \implies \{a_i, a_k\} = 0 \\ a_i^\dagger a_k^\dagger &= -a_k^\dagger a_i^\dagger \implies \{a_i^\dagger, a_k^\dagger\} = 0 \end{aligned}$$

By the Pauli Exclusion Principle, identical fermions cannot occupy the same state, meaning $a_i a_i = 0$ and $a_i^\dagger a_i^\dagger = 0$:

$$\{a_i, a_i\} = a_i a_i + a_i a_i = 0$$
$$\{a_i^\dagger, a_i^\dagger\} = a_i^\dagger a_i^\dagger + a_i^\dagger a_i^\dagger = 0$$

In summary, the fundamental fermionic anticommutation relations are:

$$\{a_i, a_k^\dagger\} = \delta_{ik} \tag{19}$$
$$\{a_i, a_k\} = \{a_i^\dagger, a_k^\dagger\} = 0$$

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